



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

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EXPERIMENT 1

EXPERIMENT 1

1. **Title:** Demonstration of Inline, Internal, and External JavaScript Execution Types & Console Methods

2. Software/Tools Required

- **Operating System:** Windows 10/11 / macOS / Linux
- **IDE/Editor:** Visual Studio Code / IntelliJ IDEA
- **Web Browser:** Google Chrome / Mozilla Firefox / Microsoft Edge
- **Languages/Tech:** HTML5, CSS3

3. Program Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Practical 1 - JS Execution & Console Methods</title>

  <link rel="stylesheet" href="styles.css">


<!-- Internal JavaScript -->

<script>

  console.info("Internal JS loaded in <head>.");

  function handleInternalClick() {

    alert("Internal JS: Executed successfully!");
```



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```
console.log("Internal JS Button Clicked.");
```

```
}
```

```
</script>
```

```
</head>
```

```
<body>
```

```
<!-- Student Profile Header Badge -->
```

```
<div class="prn-badge-container">
```

```
<div class="prn-badge">
```

```
<span>Student Name: <strong>Parth Jani</strong></span>
```

```
<span>PRN: <strong>24070521241</strong></span>
```

```
</div>
```

```
</div>
```

```
<div class="main-wrapper">
```

```
<!-- Section 1: Internal JS -->
```

```
<section class="demo-section">
```

```
<h3 class="section-title">1. Internal JS</h3>
```

```
<h1 class="heading-blue">Welcome to Internal JS</h1>
```

```
<p class="text-red">This webpage is styled and scripted using Internal JS.</p>
```

```
<button class="btn-blue" onclick="handleInternalClick()">Click Here</button>
```

```
</section>
```

```
<!-- Section 2: Inline JS -->
```

```
<section class="demo-section">
```



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<h3 class="section-title">2. Inline JS</h3>

<h1 class="heading-lime">Welcome to Inline JS</h1>

<p class="text-purple">This webpage is styled and scripted using Inline JS.</p>

<button class="btn-blue" onclick="alert('Inline JS: Triggered directly from HTML attribute!'); console.log('Inline JS triggered.');">Click Here</button>

</section>

<!-- Section 3: External JS -->

<section class="demo-section no-border">

<h3 class="section-title">3. External JS</h3>

<h1 class="heading-green">Welcome to External JS</h1>

<p class="text-purple">This webpage is styled and scripted using External JS.</p>

<button class="btn-orange" id="externalBtn">Click Here</button>

</section>

<!-- User Information & Welcome Display Card -->

<section class="user-info-card">

<div id="welcomeBanner" class="welcome-text"></div>

<div id="userInfoDisplay" class="info-grid"></div>

</section>

</div>

<!-- External JavaScript Link -->

<script src="script.js"></script>

</body>



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</html>

* {

box-sizing: border-box;

margin: 0;

padding: 0;

font-family: Arial, Helvetica, sans-serif;

}

body {

background-color: #f4f4f9;

color: #000000;

padding: 20px 0;

}

.prn-badge-container {

max-width: 900px;

margin: 0 auto 20px auto;

padding: 0 40px;

}

.prn-badge {

display: flex;

justify-content: space-between;

align-items: center;

background-color: #1e3a8a;



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color: #ffffff;

padding: 12px 20px;

border-radius: 6px;

font-size: 15px;

flex-wrap: wrap;

gap: 10px;

}

.prn-badge strong {

background: rgba(255, 255, 255, 0.2);

padding: 4px 10px;

border-radius: 4px;

letter-spacing: 0.5px;

}

.main-wrapper {

max-width: 900px;

margin: 0 auto;

padding: 10px 40px;

}

.demo-section {

padding-bottom: 25px;

margin-bottom: 25px;

border-bottom: 1px solid #e0e0eb;



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}

.demo-section.no-border {

border-bottom: none;

}

.section-title {

font-size: 20px;

font-weight: bold;

color: #000000;

margin-bottom: 16px;

}

.heading-blue { color: #0000d6; font-size: 28px; font-weight: bold; margin-bottom: 12px; }

.heading-lime { color: #66cc33; font-size: 28px; font-weight: bold; margin-bottom: 12px; }

.heading-green { color: #1e8437; font-size: 28px; font-weight: bold; margin-bottom: 12px; }

.text-red { color: #d93838; font-size: 16px; margin-bottom: 16px; }

.text-purple { color: #8a2be2; font-size: 16px; margin-bottom: 16px; }

.btn-blue {

background-color: #4a90e2;

color: #ffffff;



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```
border: none;

padding: 10px 22px;

border-radius: 6px;

font-size: 13px;

font-weight: bold;

cursor: pointer;

}

.btn-orange {

background-color: #f5a623;

color: #ffffff;

border: none;

padding: 10px 22px;

border-radius: 6px;

font-size: 13px;

font-weight: bold;

cursor: pointer;

}
```

```
.user-info-card {

background-color: #ffffff;

border: 1px solid #d0d0e0;

border-radius: 8px;

padding: 20px;

margin-top: 20px;

}
```



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}

```
.welcome-text {  
  font-size: 18px;  
  font-weight: bold;  
  color: #1e3a8a;  
  margin-bottom: 12px;  
}
```

```
.info-grid {  
  display: grid;  
  grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));  
  gap: 10px;  
  font-size: 14px;  
  color: #333333;  
}
```

```
console.log("%c--- Practical 1 Executing ---", "color: #1e8437; font-weight: bold;  
font-size: 14px;");
```

```
console.info("External JS: script.js successfully loaded.");
```

```
console.warn("Console Warning: Demonstrating console methods.");
```

```
console.error("Console Error Demo: System check completed without errors.");
```

```
const userProfile = {  
  name: "Parth Jani",  
  prn: "24070521241",
```




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status: "Active Student",

batch: "2024-2028",

semester: "Semester V"

};

console.table(userProfile);

console.group("System Diagnostic Logs");

console.log("Checking DOM loading state...");

console.log("Binding external button handlers...");

console.groupEnd();

document.addEventListener('DOMContentLoaded', () => {

const externalBtn = document.getElementById('externalBtn');

const welcomeBanner = document.getElementById('welcomeBanner');

const userInfoDisplay = document.getElementById('userInfoDisplay');

const hour = new Date().getHours();

let greeting = "Welcome";

if (hour < 12) greeting = "Good Morning";

else if (hour < 18) greeting = "Good Afternoon";

else greeting = "Good Evening";

welcomeBanner.textContent = `\${greeting}, \${userProfile.name}! System Session Initialized.`;



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```
userInfoDisplay.innerHTML = `  
  <div><strong>Student Name:</strong> ${userProfile.name}</div>  
  <div><strong>PRN:</strong> ${userProfile.prn}</div>  
  <div><strong>Status:</strong> ${userProfile.status}</div>  
  <div><strong>Batch:</strong> ${userProfile.batch}</div>  
  <div><strong>Semester:</strong> ${userProfile.semester}</div>  
`;  
;
```

```
externalBtn.addEventListener('click', () => {  
  alert(`External JS: Triggered for ${userProfile.name} from script.js file!`);  
  console.log("External JS Button Clicked.");  
});  
});
```

4. Output



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1. Internal JS

Welcome to Internal JS

This webpage is styled and scripted using Internal JS.

[Click Here](#)

2. Inline JS

Welcome to Inline JS

This webpage is styled and scripted using Inline JS.

[Click Here](#)

3. External JS

Welcome to External JS

This webpage is styled and scripted using External JS.

[Click Here](#)

Good Afternoon, Parth Jani! System Session Initialized.

Student Name: Parth Jani

PRN: 24070521241

Status: Active Student

Batch: 2024-2028

Semester: Semester V

- **File Path:**
/Users/pj/Downloads/JavaScript/Experiments/Experiment-1(24070521241)
- **Student Name:** Parth Jani
- **Student PRN:** 24070521241

5. Result/Conclusion

The execution of inline, internal, and external JavaScript integration styles alongside developer console logging functions (`console.log`, `info`, `warn`, `error`, `table`) was successfully verified for **Parth Jani** (PRN: **24070521241**)



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Case Study-1

1. Title: Interactive Academic Portal: Linking College and Student Information Systems

2. Software/Tools Required

- **Operating System:** Windows 10/11 / macOS / Linux
- **IDE/Editor:** Visual Studio Code / IntelliJ IDEA
- **Web Browser:** Google Chrome / Mozilla Firefox / Microsoft Edge
- **Languages/Tech:** HTML5, CSS3, JavaScript (ES6+)

3. Program Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Academic Portal System</title>

  <link rel="stylesheet" href="styles.css">

</head>

<body>

  <div class="container">

    <header class="badge-header full-width">

      <div><strong>STUDENT NAME:</strong> Parth Jani</div>
```



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<div>STUDENT PRN: 24070521241</div>

</header>

<section class="card">

<h2>Institutional Profile Configuration</h2>

<div class="form-group">

<label>Institution Name</label>

<input type="text" value="Symbiosis Institute of Technology" readonly>

</div>

<div class="form-group">

<label for="deptSelect">Department</label>

<select id="deptSelect">

<option value="Computer Science & Engineering">Computer Science & Engineering</option>

<option value="Information Technology">Information Technology</option>

<option value="AI & Machine Learning">AI & Machine Learning</option>

</select>

</div>

<div class="form-group">

<label for="batchSelect">Academic Batch</label>

<select id="batchSelect">

<option value="2024-2028">2024–2028</option>

<option value="2023-2027">2023–2027</option>

</select>

</div>

</section>



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<section class="card">

<h2>Student Record Validation Form</h2>

<form id="studentForm" novalidate>

<div class="form-group">

<label for="sName">Full Name</label>

<input type="text" id="sName" placeholder="e.g. Parth Jani" required>

<small class="error" id="nameErr">Name must be at least 3 letters long.</small>

</div>

<div class="form-group">

<label for="sPrn">PRN (11 Digits)</label>

<input type="text" id="sPrn" placeholder="24070521241" maxlength="11" required>

<small class="error" id="prnErr">PRN must be exactly 11 numeric digits.</small>

</div>

<div class="form-group">

<label for="sEmail">Institutional Email</label>

<input type="email" id="sEmail" placeholder="parth.jani@sit.edu.in" required>

<small class="error" id="emailErr">Enter a valid institutional email address.</small>

</div>

<button type="submit" class="btn-primary">Register Student Card</button>

</form>

</section>

<section class="card full-width">

<h2>Active Profile State</h2>



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```
<div id="cardOutput" class="profile-card-display">
```

```
    <p class="placeholder">Submit student information to link profile state.</p>
```

```
</div>
```

```
</section>
```

```
</div>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

```
* { box-sizing: border-box; margin: 0; padding: 0; font-family: 'Segoe UI', sans-serif; }
```

```
body { background: #f8fafc; color: #0f172a; padding: 20px; }
```

```
.container { max-width: 950px; margin: 0 auto; display: grid; grid-template-columns: 1fr 1fr; gap: 20px; }
```

```
.full-width { grid-column: 1 / -1; }
```

```
.badge-header { background: #1e3a8a; color: white; padding: 14px 20px; border-radius: 8px; display: flex; justify-content: space-between; }
```

```
.card { background: white; border: 1px solid #cbd5e1; padding: 20px; border-radius: 10px; }
```

```
h2 { color: #1e3a8a; font-size: 16px; border-bottom: 2px solid #2563eb; padding-bottom: 6px; margin-bottom: 14px; }
```

```
.form-group { margin-bottom: 12px; }
```

```
label { display: block; font-size: 12px; font-weight: 600; margin-bottom: 4px; }
```

```
input, select { width: 100%; padding: 8px; border: 1px solid #cbd5e1; border-radius: 6px; font-size: 13px; }
```

```
.error { color: #dc2626; font-size: 11px; display: none; margin-top: 2px; }
```

```
.btn-primary { background: #1e3a8a; color: white; border: none; padding: 10px; width: 100%; border-radius: 6px; cursor: pointer; font-weight: 600; }
```

```
.profile-card-display { background: #f1f5f9; border: 1px solid #cbd5e1; padding: 16px; border-radius: 8px; }
```



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```
const AcademicPortal = (() => {

  let activeStudent = null;

  const DOM = {

    dept: document.getElementById('deptSelect'),

    batch: document.getElementById('batchSelect'),

    form: document.getElementById('studentForm'),

    name: document.getElementById('sName'),

    prn: document.getElementById('sPrn'),

    email: document.getElementById('sEmail'),

    output: document.getElementById('cardOutput'),

    nameErr: document.getElementById('nameErr'),

    prnErr: document.getElementById('prnErr'),

    emailErr: document.getElementById('emailErr')

  };

  const validate = () => {

    let valid = true;

    if (DOM.name.value.trim().length < 3) { DOM.nameErr.style.display = 'block'; valid = false; }

    else DOM.nameErr.style.display = 'none';

    if (!/^[d{11}$/.test(DOM.prn.value.trim())) { DOM.prnErr.style.display = 'block'; valid = false; }

    else DOM.prnErr.style.display = 'none';

    if (!/^[^s@]+@[^s@]+\.[^s@]+$/ .test(DOM.email.value.trim())) {

      DOM.emailErr.style.display = 'block'; valid = false; }

  }

}
```




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```
else DOM.emailErr.style.display = 'none';
```

```
return valid;
```

```
};
```

```
const render = () => {
```

```
  if (!activeStudent) return;
```

```
  DOM.output.innerHTML = `
```

```
    <div style="border-left: 4px solid #1e3a8a; padding-left: 12px;">
```

```
      <h3>${activeStudent.name}</h3>
```

```
      <p><strong>PRN:</strong> ${activeStudent.prn}</p>
```

```
      <p><strong>Email:</strong> ${activeStudent.email}</p>
```

```
      <p><strong>Department:</strong> ${DOM.dept.value}</p>
```

```
      <p><strong>Batch:</strong> ${DOM.batch.value}</p>
```

```
      <p><strong>Institution:</strong> Symbiosis Institute of Technology</p>
```

```
    </div>
```

```
  `;
```

```
};
```

```
return {
```

```
  init: () => {
```

```
    DOM.form.addEventListener('submit', (e) => {
```

```
      e.preventDefault();
```

```
      if (validate()) {
```

```
        activeStudent = { name: DOM.name.value.trim(), prn: DOM.prn.value.trim(),  
email: DOM.email.value.trim() };
```



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```
render();

}

});

DOM.dept.addEventListener('change', render);

DOM.batch.addEventListener('change', render);

}

};

})();

document.addEventListener('DOMContentLoaded', AcademicPortal.init);
```

4. Output:

ks toolbar. [Manage bookmarks...](#)

Student PRN:24070521241

Institutional Profile Configuration

Institution	Department	Academic Batch
Symbiosis Institute of Technology	Computer Science & Engineering	2024-2028

Student Information Form

Student Full Name
Parth Jani

Permanent Registration Number (PRN)
24070521241

Institutional Email
abc@sit.edu.in

Semester
Semester V (3rd Year)

Link & Generate Profile

Linked System Output

Parth Jani
PRN: 24070521241

Institution:	Symbiosis Institute of Technology
Department:	Computer Science & Engineering
Batch:	2024-2028
Semester:	Semester V (3rd Year)
Email:	abc@sit.edu.in
System Status:	Verified & Linked

Mouza - Wathoda, Nagpur 440008, Maharashtra, India. | Tel: +91-712-6192370 / +91-712-6192368
Email: info@sitnagpur.siu.edu.in | Website: www.sitnagpur.edu.in



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- **File Path:**
/Users/pj/Downloads/JavaScript/Case-Study/CaseStudy-1
- **Student Name:** Parth Jani
- **Student PRN:** 24070521241



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Experiment:2

1. Title: Demonstration of JavaScript ES6+ Variable Declarations (var, let, const), Template Literals, and Destructuring via an Automated Billing Calculator

2. Software/Tools Required

- **Operating System:** Windows 10/11 / macOS / Linux
- **IDE/Editor:** Visual Studio Code / IntelliJ IDEA
- **Web Browser:** Google Chrome / Mozilla Firefox / Microsoft Edge
- **Languages/Tech:** HTML5, CSS3, JavaScript (ES6+)

Program Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Practical 2 - ES6 Features & Billing Calculator</title>

  <link rel="stylesheet" href="styles.css">

</head>

<body>

  <div class="app-wrapper">

    <header class="profile-header-card">
```



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```
<div class="profile-detail">

  <span class="detail-label">STUDENT NAME:</span>

  <span class="detail-value">Parth Jani</span>

</div>
```

```
<div class="profile-detail">

  <span class="detail-label">STUDENT PRN:</span>

  <span class="detail-value">24070521241</span>

</div>
```

```
</header>
```

```
<main class="grid-container">
```

```
<section class="glass-card">
```

```
<h2 class="card-title">Add Billing Item</h2>
```

```
<form id="billingForm">
```

```
<div class="input-group">
```

```
<label for="itemName">Item Name</label>
```

```
<input type="text" id="itemName" placeholder="e.g. Mechanical Keyboard"
required>
```

```
</div>
```

```
<div class="input-row">
```

```
<div class="input-group">
```

```
<label for="itemPrice">Price (₹)</label>
```

```
<input type="number" id="itemPrice" min="0.01" step="0.01"
placeholder="3500" required>
```

```
</div>
```



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```
<div class="input-group">
```

```
<label for="itemQty">Quantity</label>
```

```
<input type="number" id="itemQty" min="1" value="1" required>
```

```
</div>
```

```
</div>
```

```
<button type="submit" class="accent-btn">Add to Invoice</button>
```

```
</form>
```

```
</section>
```

```
<section class="glass-card">
```

```
<h2 class="card-title">Live Calculations</h2>
```

```
<div id="summaryDisplay" class="summary-box">
```

```
<p class="empty-notice">No items added to invoice yet.</p>
```

```
</div>
```

```
</section>
```

```
<section class="glass-card full-width">
```

```
<div class="table-header">
```

```
<h2 class="card-title">Itemized Invoice Items</h2>
```

```
<button id="clearAllBtn" class="clear-btn">Clear All</button>
```

```
</div>
```

```
<div class="table-wrapper">
```

```
<table class="custom-table">
```

```
<thead>
```



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<tr>

<th>#</th>

<th>Item Name</th>

<th>Unit Price</th>

<th>Qty</th>

<th>Subtotal</th>

<th>Action</th>

</tr>

</thead>

<tbody id="cartItemsBody"></tbody>

</table>

</div>

</section>

</main>

</div>

<script src="script.js"></script>

</body>

</html>

:root {

--primary: #0f766e;

--primary-hover: #115e59;

--secondary: #14b8a6;

--card-bg: rgba(255, 255, 255, 0.9);

--text-dark: #0f172a;

--text-muted: #64748b;



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```
--border-color: #cbd5e1;

--danger: #ef4444;

}

* { box-sizing: border-box; margin: 0; padding: 0; font-family: 'Segoe UI', sans-serif; }

body { background: linear-gradient(135deg, #ecfdf5 0%, #e0f2fe 100%); padding: 24px; }

.app-wrapper { max-width: 1050px; margin: 0 auto; }

.profile-header-card {

  display: flex; justify-content: space-between; margin-bottom: 24px; padding: 16px 24px;

  background: #064e3b; color: white; border-radius: 12px;

}

.profile-detail { display: flex; align-items: center; gap: 8px; }

.detail-label { font-size: 12px; font-weight: 600; color: #14b8a6; }

.detail-value { font-size: 18px; font-weight: 700; background: rgba(255, 255, 255, 0.15);

padding: 4px 14px; border-radius: 6px; }

.grid-container { display: grid; grid-template-columns: 1fr 1fr; gap: 20px; }

.glass-card { background: var(--card-bg); border-radius: 14px; padding: 24px; border: 1px

solid white; }

.full-width { grid-column: 1 / -1; }

.card-title { font-size: 18px; color: var(--primary); margin-bottom: 18px; font-weight: 700;

border-left: 4px solid var(--secondary); padding-left: 10px; }

.input-group { margin-bottom: 14px; }

.input-row { display: grid; grid-template-columns: 1fr 1fr; gap: 12px; }

label { display: block; font-size: 13px; font-weight: 600; color: var(--text-muted);

margin-bottom: 6px; }
```




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```
input { width: 100%; padding: 10px; border: 1px solid var(--border-color); border-radius: 8px; }
```

```
.accent-btn { width: 100%; background: var(--primary); color: white; border: none; padding: 12px; border-radius: 8px; font-weight: 600; cursor: pointer; }
```

```
.summary-box { background: #f0fdf4; border: 1px solid #bbf7d0; border-radius: 10px; padding: 16px; }
```

```
.summary-line { display: flex; justify-content: space-between; margin-bottom: 10px; font-size: 14px; }
```

```
.summary-line.grand-total { font-size: 18px; font-weight: 700; color: var(--primary); border-top: 2px dashed #99f6e4; padding-top: 10px; }
```

```
.table-header { display: flex; justify-content: space-between; align-items: center; }
```

```
.clear-btn { background: transparent; border: 1px solid var(--danger); color: var(--danger); padding: 6px 12px; border-radius: 6px; cursor: pointer; }
```

```
.custom-table { width: 100%; border-collapse: collapse; margin-top: 12px; }
```

```
.custom-table th { background: #ccfbf1; color: var(--primary); text-align: left; padding: 12px; font-size: 13px; }
```

```
.custom-table td { padding: 12px; border-bottom: 1px solid #f1f5f9; font-size: 14px; }
```

```
.del-item-btn { background: #fee2e2; color: var(--danger); border: none; padding: 4px 8px; border-radius: 4px; cursor: pointer; }
```

```
var GLOBAL_TAX_RATE = 0.18;
```

```
const STUDENT_NAME = "Parth Jani";
```

```
const PRN_ID = "24070521241";
```

```
const DISCOUNT_THRESHOLD = 3000;
```

```
let billingCart = [];
```

```
const DOM = {
```



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```
form: document.getElementById('billingForm'),

nameInput: document.getElementById('itemName'),

priceInput: document.getElementById('itemPrice'),

qtyInput: document.getElementById('itemQty'),

summaryDisplay: document.getElementById('summaryDisplay'),

cartBody: document.getElementById('cartItemsBody'),

clearBtn: document.getElementById('clearAllBtn')

};

const calculateInvoiceTotals = () => {

  let subtotal = 0;

  billingCart.forEach(item => {

    const { price, qty } = item;

    subtotal += price * qty;

  });

  let discount = subtotal >= DISCOUNT_THRESHOLD ? subtotal * 0.10 : 0;

  let taxableAmount = subtotal - discount;

  let taxAmount = taxableAmount * GLOBAL_TAX_RATE;

  let grandTotal = taxableAmount + taxAmount;

  return [subtotal, discount, taxAmount, grandTotal];

};

const renderApp = () => {

  DOM.cartBody.innerHTML = '';
```



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```
if (billingCart.length === 0) {
```

```
    DOM.cartBody.innerHTML = `<tr><td colspan="6" style="text-align:center; padding:
20px;">No items added yet.</td></tr>`;
```

```
    DOM.summaryDisplay.innerHTML = `<p class="empty-notice">No items added to
invoice yet.</p>`;
```

```
    return;
```

```
}
```

```
billingCart.forEach((item, index) => {
```

```
    const { name, price, qty } = item;
```

```
    const itemSubtotal = price * qty;
```

```
    DOM.cartBody.innerHTML += `
```

```
        <tr>
```

```
            <td><strong>${index + 1}</strong></td>
```

```
            <td>${name}</td>
```

```
            <td>₹${price.toFixed(2)}</td>
```

```
            <td>${qty}</td>
```

```
            <td>₹${itemSubtotal.toFixed(2)}</td>
```

```
            <td><button class="del-item-btn"
onclick="removeBillingItem(${index})">Remove</button></td>
```

```
        </tr>
```

```
    `;
```

```
});
```

```
const [subtotal, discount, taxAmount, grandTotal] = calculateInvoiceTotals();
```



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```
DOM.summaryDisplay.innerHTML = `
```

```
    <div class="summary-line"><span>Student Billed:</span>
    <span><strong>${STUDENT_NAME} (${PRN_ID})</strong></span></div>
```

```
    <div class="summary-line"><span>Subtotal:</span>
    <span>₹${subtotal.toFixed(2)}</span></div>
```

```
    <div class="summary-line"><span>Discount (10% over ₹3k):</span>
    <span>-₹${discount.toFixed(2)}</span></div>
```

```
    <div class="summary-line"><span>GST Rate (18%):</span>
    <span>+₹${taxAmount.toFixed(2)}</span></div>
```

```
    <div class="summary-line grand-total"><span>Grand Total:</span>
    <span>₹${grandTotal.toFixed(2)}</span></div>
```

```
`;
```

```
};
```

```
DOM.form.addEventListener('submit', (e) => {
```

```
    e.preventDefault();
```

```
    let name = DOM.nameInput.value.trim();
```

```
    let price = parseFloat(DOM.priceInput.value);
```

```
    let qty = parseInt(DOM.qtyInput.value);
```

```
    if (name && !isNaN(price) && !isNaN(qty)) {
```

```
        billingCart.push({ name, price, qty });
```

```
        DOM.form.reset();
```

```
        DOM.qtyInput.value = 1;
```

```
        renderApp();
```

```
    }
```

```
});
```



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```
window.removeBillingItem = (index) => { billingCart.splice(index, 1); renderApp(); };
```

```
DOM.clearBtn.addEventListener('click', () => { billingCart = []; renderApp(); });
```

```
document.addEventListener('DOMContentLoaded', renderApp);
```

4. Output:

PRN 24070521241

Billing Calculator & ES6 Engine

Add Billing Item

Item Name

e.g. Mechanical Keyboard

Price (₹)

1500

Quantity

1

Add to Invoice

Live Calculations

Subtotal:

₹3500.00

Discount (10% over ₹3,000):

-₹350.00

GST Rate (18%):

+₹567.00

Grand Total:

₹3717.00

Itemized Invoice Items

Clear All

#	Item Name	Unit Price	Qty	Subtotal	Action
1	Mouse	₹1500.00	1	₹1500.00	Remove
2	Keyobard	₹2000.00	1	₹2000.00	Remove

- **File Path:**

/Users/pj/Downloads/JavaScript/Experiments/Experiment-2(24070521241)

- **Student Name:** Parth Jani

- **Student PRN:** 24070521241



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5. Result/Conclusion

Core ES6 features including scoping (**let**, **const**), template literals, and array/object destructuring were successfully implemented inside an automated billing engine for **Parth Jani** (PRN: **24070521241**).



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Case Study: 2

1. Title: Automated E-Commerce Billing Engine & Dynamic Invoice Generator

Software/Tools Required

- **Operating System:** Windows 10/11 / macOS / Linux
- **IDE/Editor:** Visual Studio Code / IntelliJ IDEA
- **Web Browser:** Google Chrome / Mozilla Firefox / Microsoft Edge
- **Languages/Tech:** HTML5, CSS3, JavaScript (ES6+)

Program Code:

bill.html

HTML

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>E-Commerce Billing Engine</title>
```

```
<link rel="stylesheet" href="styles.css">
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<header class="badge-header full-width">
```



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STUDENT NAME: Parth Jani

PRN: 24070521241

</header>

<section class="card">

<h2>Product Selection</h2>

<form id="cartForm">

<div class="form-group">

<label for="productSelect">Select Product</label>

<select id="productSelect">

<option value="Laptop" data-price="65000">Laptop (₹65,000)</option>

<option value="Wireless Mouse" data-price="1200">Wireless Mouse
(₹1,200)</option>

<option value="Monitor" data-price="15000">Monitor (₹15,000)</option>

</select>

</div>

<div class="form-group">

<label for="qty">Quantity</label>

<input type="number" id="qty" min="1" value="1">

</div>

<button type="submit" class="btn-primary">Add Item To Invoice</button>

</form>

</section>

<section class="card">

<h2>Tax & Discount Settings</h2>



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```
<div class="form-group">

  <label for="gstInput">GST Rate (%)</label>

  <input type="number" id="gstInput" value="18" min="0">

</div>

</section>
```

```
<section class="card full-width">

  <h2>Generated Tax Invoice</h2>

  <div id="invoiceOutput">

    <p class="placeholder">Add products to generate invoice state.</p>

  </div>

</section>

</div>

<script src="script.js"></script>

</body>

</html>
```

styles.css

```
* { box-sizing: border-box; margin: 0; padding: 0; font-family: 'Segoe UI', sans-serif; }

body { background: #f0fdf4; color: #064e3b; padding: 20px; }

.container { max-width: 900px; margin: 0 auto; display: grid; grid-template-columns: 1fr 1fr; gap: 20px; }

.full-width { grid-column: 1 / -1; }

.badge-header { background: #064e3b; color: white; padding: 14px 20px; border-radius: 8px; display: flex; justify-content: space-between; }

.card { background: white; border: 1px solid #a7f3d0; padding: 20px; border-radius: 10px; }
```



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h2 { color: #064e3b; font-size: 16px; border-bottom: 2px solid #10b981; padding-bottom: 6px; margin-bottom: 14px; }

.form-group { margin-bottom: 12px; }

label { display: block; font-size: 12px; font-weight: 600; margin-bottom: 4px; }

input, select { width: 100%; padding: 8px; border: 1px solid #cbd5e1; border-radius: 6px; }

.btn-primary { background: #064e3b; color: white; border: none; padding: 10px; width: 100%; border-radius: 6px; cursor: pointer; }

.invoice-box { background: #ffffff; border: 2px solid #064e3b; padding: 16px; border-radius: 8px; }

script.js

```
const BillingApp = (() => {  
  let items = [];  
  const DOM = {  
    form: document.getElementById('cartForm'),  
    product: document.getElementById('productSelect'),  
    qty: document.getElementById('qty'),  
    gst: document.getElementById('gstInput'),  
    output: document.getElementById('invoiceOutput')  
  };  
};
```

```
const calculateTotals = () => {  
  let subtotal = items.reduce((acc, i) => acc +  
(i.price * i.qty), 0);  
  let discountRate = 0;  
  if (subtotal > 20000) discountRate = 0.20;  
  else if (subtotal > 10000) discountRate = 0.15;
```



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```
else if (subtotal > 5000) discountRate = 0.10;

let discount = subtotal * discountRate;

let taxable = subtotal - discount;

let tax = taxable * (parseFloat(DOM.gst.value || 18)
/ 100);

let grandTotal = taxable + tax;

return { subtotal, discount, tax, grandTotal };

};

const render = () => {

  if (items.length === 0) return;

  const { subtotal, discount, tax, grandTotal } =
calculateTotals();

  DOM.output.innerHTML = `

    <div class="invoice-box">

      <h3>TAX INVOICE - Ref
#${Math.floor(Math.random() * 90000 + 10000)}</h3>

      <p><strong>Billed Student:</strong> Parth
Jani (PRN: 24070521241)</p>

      <hr style="margin:10px 0;">

      <p>Subtotal: ₹${subtotal.toFixed(2)}</p>

      <p>Discount Reduction:
-₹${discount.toFixed(2)}</p>

      <p>GST Tax Addition: +₹${tax.toFixed(2)}</p>

      <h4 style="margin-top:10px;
```



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color:#064e3b;">Grand Total: ₹\${grandTotal.toFixed(2)}</h4>

</div>

`;

};

return {

init: () => {

DOM.form.addEventListener('submit', (e) => {

e.preventDefault();

const sel =

DOM.product.options[DOM.product.selectedIndex];

items.push({ name: sel.value, price:
parseFloat(sel.dataset.price), qty: parseInt(DOM.qty.value)
});

render();

});

DOM.gst.addEventListener('input', render);

}

};

})();

document.addEventListener('DOMContentLoaded',
BillingApp.init);



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4. Output:

Student PRN: 24070521241

Product Selection & Cart

Select Item

Choose a product...

Quantity

1

Add to Cart

Item Description	Price	Qty	Total	Action
Mechanical Keyboard	₹3500.00	1	₹3500.00	Remove
USB-C Docking Station	₹2800.00	3	₹8400.00	Remove

Billing Calculation Settings

Applicable Tax Rate (GST %)

18

Automatic Discount Thresholds

- Subtotal > ₹5,000 : **10% Discount**
- Subtotal > ₹10,000 : **15% Discount**
- Subtotal > ₹20,000 : **20% Discount**

Generate Itemized Invoice

Rendered Itemized Invoice

TAX INVOICE
Ref: INV-340076

Date: 8/6/2026
PRN: 24070521241

Item Name	Unit Price	Qty	Total
Mechanical Keyboard	₹3500.00	1	₹3500.00
USB-C Docking Station	₹2800.00	3	₹8400.00

Subtotal:

₹11900.00

Discount (15%):

-₹1785.00

GST (18%):

+₹1820.70

Grand Total:

₹11935.70

- **File Path:**
/Users/pj/Downloads/JavaScript/Case-Study/CaseStudy-2
- **Student Name:** Parth Jani
- **Student PRN:** 24070521241

5. Result/Conclusion

The dynamic invoice generator parsed real-time user selections, applied tax and tier-based discount logic, and rendered structured tax invoices for **Parth Jani** (PRN: **24070521241**).



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Experiment 3

1. **Title:** Implementation of Client-Side Form Validation and JavaScript Control Structures (if-else, switch, loops) via an Academic Grade Evaluation System

2. Software/Tools Required

- **Operating System:** Windows 10/11 / macOS / Linux
- **IDE/Editor:** Visual Studio Code / IntelliJ IDEA
- **Web Browser:** Google Chrome / Mozilla Firefox / Microsoft Edge
- **Languages/Tech:** HTML5, CSS3, JavaScript (ES6+)

Program Code

Index.html

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width,
initial-scale=1.0">

    <title>Practical 3 - Grade Evaluation System</title>

    <link rel="stylesheet" href="styles.css">

</head>

<body>

<div class="container">
```



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```
<div class="profile-header-card full-width">

  <div class="profile-detail">

    <span class="detail-label">STUDENT NAME:</span>

    <span class="detail-value">Parth Jani</span>

  </div>

  <div class="profile-detail">

    <span class="detail-label">STUDENT PRN:</span>

    <span class="detail-value">24070521241</span>

  </div>

</div>

<section class="card">

  <h2>Marks Evaluation Form</h2>

  <form id="gradeForm" novalidate>

    <div class="form-group">

      <label for="sub1">Web Technologies (0-100)</label>

      <input type="number" id="sub1" class="mark-input"
min="0" max="100" placeholder="e.g. 85" required>

      <div class="error-msg">Please enter valid marks
between 0 and 100.</div>

    </div>

    <div class="form-group">

      <label for="sub2">Data Structures (0-100)</label>

      <input type="number" id="sub2" class="mark-input"
min="0" max="100" placeholder="e.g. 78" required>

      <div class="error-msg">Please enter valid marks
between 0 and 100.</div>

    </div>

  </form>

</section>
```



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</div>

<div class="form-group">

<label for="sub3">Database Management Systems
(0-100)</label>

<input type="number" id="sub3" class="mark-input"
min="0" max="100" placeholder="e.g. 92" required>

<div class="error-msg">Please enter valid marks
between 0 and 100.</div>

</div>

<div class="form-group">

<label for="sub4">Computer Networks
(0-100)</label>

<input type="number" id="sub4" class="mark-input"
min="0" max="100" placeholder="e.g. 64" required>

<div class="error-msg">Please enter valid marks
between 0 and 100.</div>

</div>

<div class="form-group">

<label for="sub5">Operating Systems
(0-100)</label>

<input type="number" id="sub5" class="mark-input"
min="0" max="100" placeholder="e.g. 88" required>

<div class="error-msg">Please enter valid marks
between 0 and 100.</div>

</div>

<button type="submit" class="btn-primary">Calculate
Grades</button>



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</form>

</section>

<section class="card">

<h2>Academic Performance Sheet</h2>

<div id="gradeOutput">

<p class="placeholder-text">Enter subject marks and submit form.</p>

</div>

</section>

</div>

<script src="script.js"></script>

</body>

</html>

styles.css

CSS

:root {

--primary: #4f46e5;

--primary-hover: #4338ca;

--secondary: #6366f1;

--bg: #f5f3ff;

--card-bg: #ffffff;

--text: #1e1b4b;

--error: #dc2626;

--border: #e0e7ff;



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}

```
* { box-sizing: border-box; margin: 0; padding: 0; font-family:
'Segoe UI', sans-serif; }
```

```
body { background-color: var(--bg); color: var(--text); padding:
20px; }
```

```
.container { max-width: 1000px; margin: 0 auto; display: grid;
grid-template-columns: 1fr 1fr; gap: 20px; }
```

```
.card { background: var(--card-bg); padding: 24px; border-radius:
12px; }
```

```
.full-width { grid-column: 1 / -1; }
```

```
.profile-header-card {
```

```
    background: var(--primary); color: white; padding: 16px 24px;
border-radius: 8px;
```

```
    display: flex; justify-content: space-between; align-items:
center;
```

```
}
```

```
.detail-value { font-size: 18px; font-weight: 700; background:
rgba(255,255,255,0.18); padding: 4px 12px; border-radius: 4px; }
```

```
.form-group { margin-bottom: 14px; }
```

```
label { display: block; margin-bottom: 6px; font-size: 13px;
font-weight: 600; }
```

```
input { width: 100%; padding: 10px; border: 1px solid
var(--border); border-radius: 6px; }
```

```
.error-msg { color: var(--error); font-size: 12px; margin-top:
4px; display: none; }
```

```
.btn-primary { background-color: var(--primary); color: white;
border: none; padding: 12px; border-radius: 6px; cursor: pointer; }
```



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width: 100%; font-weight: 600; }

```
.result-box { background: #f8fafc; border: 1px solid  
var(--border); padding: 16px; border-radius: 8px; }
```

```
.subject-table { width: 100%; border-collapse: collapse;  
margin-bottom: 16px; }
```

```
.subject-table th, .subject-table td { padding: 8px 10px;  
text-align: left; font-size: 13px; border-bottom: 1px solid  
#e2e8f0; }
```

```
.subject-table th { background-color: #e0e7ff; color:  
var(--primary); }
```

```
.summary-row { display: flex; justify-content: space-between;  
margin-bottom: 8px; font-size: 14px; }
```

```
.summary-row.total-row { font-weight: bold; font-size: 16px;  
color: var(--primary); }
```

script.js

JavaScript

```
const GradeCalculator = (() => {  
  const subjects = [  
    { id: 'sub1', name: 'Web Technologies' },  
    { id: 'sub2', name: 'Data Structures' },  
    { id: 'sub3', name: 'Database Management Systems' },  
    { id: 'sub4', name: 'Computer Networks' },  
    { id: 'sub5', name: 'Operating Systems' }  
  ];  
  
  const DOM = {  
    form: document.getElementById('gradeForm'),
```



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```
output: document.getElementById('gradeOutput')

});

const validateForm = () => {
  let isValid = true;
  for (let sub of subjects) {
    const inputEl = document.getElementById(sub.id);
    const errorEl = inputEl.nextElementSibling;
    const val = parseFloat(inputEl.value);

    if (isNaN(val) || val < 0 || val > 100) {
      errorEl.style.display = 'block';
      isValid = false;
    } else {
      errorEl.style.display = 'none';
    }
  }
  return isValid;
};

const getSubjectGrade = (marks) => {
  if (marks >= 90) return { grade: 'O', label: 'Outstanding' };

  else if (marks >= 80) return { grade: 'A+', label: 'Excellent' };

  else if (marks >= 70) return { grade: 'A', label: 'Very Good' };
};
```



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```
    else if (marks >= 60) return { grade: 'B+', label: 'Good'
};

    else if (marks >= 50) return { grade: 'B', label: 'Above
Average' };

    else if (marks >= 40) return { grade: 'P', label: 'Pass'
};

    else return { grade: 'F', label: 'Fail' };

};
```

```
const getAcademicClassification = (percentage, hasFailed) => {
    if (hasFailed) return "Failed (Re-examination Required)";
    const scoreBracket = Math.floor((percentage / 10));

    switch (scoreBracket) {
        case 10: case 9: case 8: return "First Class with
Distinction";
        case 7: return "First Class";
        case 6: return "Higher Second Class";
        case 5: return "Second Class";
        case 4: return "Pass Class";
        default: return "Unclassified";
    }
};
```

```
const renderGradeSheet = () => {
    let totalMarks = 0;
    let hasFailed = false;
    let tableRowsHtml = '';
```



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```
subjects.forEach(sub => {

    const marks =
parseFloat(document.getElementById(sub.id).value);

    const gradeInfo = getSubjectGrade(marks);

    totalMarks += marks;

    if (gradeInfo.grade === 'F') hasFailed = true;


    tableRowsHtml += `<tr><td>${sub.name}</td><td>${marks}
/ 100</td><td>${gradeInfo.grade} (${gradeInfo.label})</td></tr>`;

});


const percentage = (totalMarks / 500) * 100;

const classification =
getAcademicClassification(percentage, hasFailed);


DOM.output.innerHTML = `

    <div class="result-box">

        <table class="subject-table">

<thead><tr><th>Subject</th><th>Marks</th><th>Grade</th></tr></thead>

        <tbody>${tableRowsHtml}</tbody>

    </table>

    <div class="summary-block">

        <div class="summary-row"><span>Student
Name:</span> <span><strong>Parth Jani</strong></span></div>

        <div class="summary-row"><span>Student
PRN:</span> <span><strong>24070521241</strong></span></div>

    </div>

</div>`
```



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```
<div class="summary-row"><span>Total
Score:</span> <span>${totalMarks} / 500</span></div>

<div
class="summary-row"><span>Percentage:</span>
<span>${percentage.toFixed(2)}%</span></div>

<div class="summary-row
total-row"><span>Result:</span> <span>${hasFailed ? 'FAIL' :
'PASS'}</span></div>

<div
class="summary-row"><span>Classification:</span>
<span><strong>${classification}</strong></span></div>

</div>

</div>

`;

};

return {

  init: () => {

    DOM.form.addEventListener('submit', (e) => {

      e.preventDefault();

      if (validateForm()) renderGradeSheet();

    });

  }

};

})();

document.addEventListener('DOMContentLoaded',
GradeCalculator.init);
```



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Output:

Student PRN: 24070521241

Marks Evaluation Form

Web Technologies (0-100)
56

Data Structures (0-100)
56

Database Management Systems (0-100)
34

Computer Networks (0-100)
75

Operating Systems (0-100)
100

Calculate Grades

Academic Performance Sheet

Subject	Marks	Grade
Web Technologies	56 / 100	B (Above Average)
Data Structures	56 / 100	B (Above Average)
Database Management Systems	34 / 100	F (Fail)
Computer Networks	75 / 100	A (Very Good)
Operating Systems	100 / 100	O (Outstanding)

Total Score: 321 / 500
Percentage: 64.20%
Result: FAIL
Classification: Failed (Re-examination Required)

- **File Path:**
/Users/pj/Downloads/JavaScript/Experiments
/Experiment-3(24070521241)
- **Student Name:** Parth Jani
- **Student PRN:** 24070521241



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Case Study-3

1. Title: Automated Academic Scholarship Evaluation & Eligibility Matrix Engine

2. Software/Tools Required

- **Operating System:** Windows 10/11 / macOS / Linux
- **IDE/Editor:** Visual Studio Code / IntelliJ IDEA
- **Web Browser:** Google Chrome / Mozilla Firefox / Microsoft Edge
- **Languages/Tech:** HTML5, CSS3, JavaScript (ES6+)

3. Program Code

ScholarshipEligibility.html

1HTML

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Academic Scholarship Eligibility Engine</title>
```

```
  <link rel="stylesheet" href="styles.css">
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
  <header class="header-block full-width">
```

```
    <div class="prn-box">
```

```
      <span class="prn-label">STUDENT PRN:</span>
```



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24070521241

</div>

<h1 class="header-title">Academic Scholarship Evaluation Engine</h1>

</header>

<section class="card">

<h2>Applicant Criteria Form</h2>

<form id="scholarshipForm" novalidate>

<div class="form-group">

<label for="cgpa">Academic Score (CGPA / 10.0)</label>

<input type="number" id="cgpa" min="0" max="10" step="0.01" placeholder="e.g. 8.75" required>

<div class="error-msg" id="cgpaError">Enter a valid CGPA between 0.00 and 10.00</div>

</div>

<div class="form-group">

<label for="income">Annual Family Income (₹)</label>

<input type="number" id="income" min="0" placeholder="e.g. 450000" required>

<div class="error-msg" id="incomeError">Enter a valid income amount.</div>

</div>

<div class="form-group">

<label for="category">Category / Demographic Group</label>

<select id="category" required>

<option value="General">General</option>

<option value="OBC">OBC (Other Backward Classes)</option>

<option value="SC/ST">SC / ST</option>

<option value="Single Girl Child">Single Girl Child</option>



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</select>

</div>

<div class="form-group">

<label for="specialQuota">Special Quota / Reservation</label>

<select id="specialQuota">

<option value="None">None</option>

<option value="Pwd">Person with Disability (PwD)</option>

<option value="Sports">National/State Level Sports</option>

</select>

</div>

<button type="submit" class="btn-submit">Evaluate Eligibility</button>

</form>

</section>

<section class="card">

<h2>Eligibility Matrix Output</h2>

<div id="evaluationOutput">

<p class="placeholder-text">Submit parameters to generate validation matrix state.</p>

</div>

</section>

</div>

<script src="script.js"></script>

</body>

</html>



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styles.css

CSS

:root {

--text-primary: #111111;

--text-secondary: #444444;

--bg-main: #fcfcfc;

--card-bg: #ffffff;

--border-dark: #111111;

--box-bg-gray: #f4f4f4;

}

* { box-sizing: border-box; margin: 0; padding: 0; font-family: 'Times New Roman', serif; }

body { background-color: var(--bg-main); color: var(--text-primary); padding: 30px; }

.container { max-width: 950px; margin: 0 auto; display: grid; grid-template-columns: 1fr 1fr; gap: 24px; }

.full-width { grid-column: 1 / -1; }

.header-block { border: 2px solid var(--border-dark); padding: 16px 20px; background: var(--card-bg); display: flex; justify-content: space-between; align-items: center; }

.header-title { font-size: 18px; font-weight: bold; text-transform: uppercase; }

.prn-box { border: 1px solid var(--border-dark); padding: 6px 12px; background: var(--box-bg-gray); font-size: 14px; }



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```
.card { border: 1px solid var(--border-dark); background: var(--card-bg); padding: 20px; }
```

```
h2 { font-size: 15px; text-transform: uppercase; border-bottom: 2px solid  
var(--border-dark); padding-bottom: 8px; margin-bottom: 16px; }
```

```
.form-group { margin-bottom: 14px; }
```

```
label { display: block; font-size: 13px; font-weight: bold; margin-bottom: 4px; }
```

```
input, select { width: 100%; padding: 8px; border: 1px solid var(--border-dark);  
background: #ffffff; font-size: 13px; }
```

```
.error-msg { color: #000; font-size: 11px; font-weight: bold; margin-top: 3px; display: none;  
}
```

```
.btn-submit { width: 100%; background: var(--border-dark); color: white; border: none;  
padding: 10px; font-weight: bold; cursor: pointer; text-transform: uppercase; }
```

```
.matrix-card { border: 2px solid var(--border-dark); padding: 16px; background:  
var(--box-bg-gray); }
```

```
.result-header { font-size: 16px; font-weight: bold; text-transform: uppercase;  
border-bottom: 1px solid var(--border-dark); padding-bottom: 6px; margin-bottom: 12px;  
}
```

```
.result-row { display: flex; justify-content: space-between; font-size: 13px; margin-bottom:  
6px; }
```

```
.divider { border-top: 1px dashed var(--border-dark); margin: 10px 0; }
```

script.js

JavaScript

```
const ScholarshipEngine = (() => {  
  
  const DOM = {  
  
    form: document.getElementById('scholarshipForm'),  
  
    cgpaInput: document.getElementById('cgpa'),  
  
    incomeInput: document.getElementById('income'),  
  
    categorySelect: document.getElementById('category'),
```



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```
quotaSelect: document.getElementById('specialQuota'),

outputBox: document.getElementById('evaluationOutput'),

cgpaErr: document.getElementById('cgpaError'),

incomeErr: document.getElementById('incomeError')

};

const validateInputs = (cgpa, income) => {

    let isValid = true;

    if (isNaN(cgpa) || cgpa < 0 || cgpa > 10) { DOM.cgpaErr.style.display = 'block'; isValid = false; }

    else DOM.cgpaErr.style.display = 'none';

    if (isNaN(income) || income < 0) { DOM.incomeErr.style.display = 'block'; isValid = false; }

    else DOM.incomeErr.style.display = 'none';

    return isValid;

};

const evaluateCriteria = (cgpa, income, category, quota) => {

    if ((cgpa >= 9.0 && income <= 300000) || (cgpa >= 8.5 && quota === 'PwD') || (cgpa >= 8.5 && category === 'SC/ST' && income <= 500000)) {

        return { status: 'APPROVED', title: 'Presidential Full Merit Scholarship', waiver: '100% Tuition Fee Waiver', stipend: '₹5,000 / Month Allowance' };

    }

    if ((cgpa >= 8.0 && income <= 600000) || (cgpa >= 7.5 && (category === 'OBC' || category === 'Single Girl Child') && income <= 400000) || (quota === 'Sports' && cgpa >= 7.5)) {

        return { status: 'APPROVED', title: 'Merit-Cum-Means Scholarship', waiver: '50%
```



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```
Tuition Fee Waiver', stipend: 'Standard Library Access' };
```

```
    }
```

```
    if (cgpa >= 7.5 && income <= 800000) {
```

```
        return { status: 'PARTIAL APPROVAL', title: 'Institutional Academic Grant', waiver:
```

```
'25% Tuition Fee Waiver', stipend: 'None' };
```

```
    }
```

```
    return { status: 'REJECTED / INELIGIBLE', title: 'Criteria Not Met', waiver: '0% Waiver',
```

```
stipend: 'None' };
```

```
};
```

```
const renderResult = (result, cgpa, income, category, quota) => {
```

```
    DOM.outputBox.innerHTML = `
```

```
        <div class="matrix-card">
```

```
            <div class="result-header">${result.title}</div>
```

```
            <div class="result-row"><span>Status State:</span> <span>[ ${result.status}
```

```
]</span></div>
```

```
            <div class="result-row"><span>Financial Benefit:</span>
```

```
<span>${result.waiver}</span></div>
```

```
            <div class="result-row"><span>Stipend / Support:</span>
```

```
<span>${result.stipend}</span></div>
```

```
            <div class="divider"></div>
```

```
            <div class="result-row"><span>Evaluated CGPA:</span>
```

```
<span>${cgpa.toFixed(2)} / 10.00</span></div>
```

```
            <div class="result-row"><span>Evaluated Income:</span>
```

```
<span>₹${income.toLocaleString('en-IN')}</span></div>
```

```
            <div class="result-row"><span>Category:</span>
```

```
<span>${category}</span></div>
```

```
            <div class="result-row"><span>Quota Applied:</span>
```

```
<span>${quota}</span></div>
```

```
        </div>
```



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```
    `;  
    };  
  
    return {  
      init: () => {  
        DOM.form.addEventListener('submit', (e) => {  
          e.preventDefault();  
  
          const cgpa = parseFloat(DOM.cgpaInput.value);  
  
          const income = parseFloat(DOM.incomeInput.value);  
  
          const category = DOM.categorySelect.value;  
  
          const quota = DOM.quotaSelect.value;  
  
          if (validateInputs(cgpa, income)) {  
            renderResult(evaluateCriteria(cgpa, income, category, quota), cgpa, income,  
category, quota);  
          }  
        });  
      }  
    };  
  }()  
  
document.addEventListener('DOMContentLoaded', ScholarshipEngine.init);
```




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4. Output:

your bookmarks here on the bookmarks toolbar. [Manage bookmarks...](#)

Scholarship Application Criteria

Academic Score (CGPA / 10)

6.76

Annual Family Income (₹)

500000

Category / Demographic Group

OBC (Other Backward Classes)

Special Quota / Reservation

Person with Disability (PWD)

Evaluate Eligibility

Eligibility Matrix Result

Criteria Not Met

Status State:	Ineligible
Financial Benefit:	0% Waiver
Stipend / Support:	None
Evaluated CGPA:	6.76 / 10
Evaluated Income:	₹5,00,000
Category:	OBC

- File Path:
/Users/pj/Downloads/JavaScript/Case-Study/CaseStudy-3
- Student Name: Parth Jani
- Student PRN: 24070521241



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5. Result/Conclusion

The Decision Logic Engine successfully processed multi-variable inputs (CGPA, income, demographics) to render formatted evaluation states for Parth Jani (PRN: 24070521241).



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Experiment-4

1. Title: Demonstration of JavaScript Function Types, Scope, Closures, and Exception Handling (try...catch) via an Interactive Palindrome Checker Application

2. Software/Tools Required

- **Operating System:** Windows 10/11 / macOS / Linux
- **IDE/Editor:** Visual Studio Code / IntelliJ IDEA
- **Web Browser:** Google Chrome / Mozilla Firefox / Microsoft Edge
- **Languages/Tech:** HTML5, CSS3, JavaScript (ES6+)

3. Program Code:

Index.html

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Practical 4 - Palindrome Checker & JS Core Mechanics</title>

  <link rel="stylesheet" href="styles.css">

</head>

<body>

<div class="container">

  <div class="profile-header-card full-width">
```



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```
<div class="profile-detail">
```

```
  <span class="detail-label">STUDENT NAME:</span>
```

```
  <span class="detail-value">Parth Jani</span>
```

```
</div>
```

```
<div class="profile-detail">
```

```
  <span class="detail-label">STUDENT PRN:</span>
```

```
  <span class="detail-value">24070521241</span>
```

```
</div>
```

```
</div>
```

```
<section class="card">
```

```
  <h2>Palindrome Evaluator</h2>
```

```
  <form id="palindromeForm" novalidate>
```

```
    <div class="form-group">
```

```
      <label for="textInput">Enter Phrase or Word</label>
```

```
      <input type="text" id="textInput" placeholder="e.g. A man, a plan, a canal: Panama" required>
```

```
    </div>
```

```
  <div class="options-group">
```

```
    <label class="checkbox-label">
```

```
      <input type="checkbox" id="ignoreSpace" checked> Ignore Spaces & Punctuation
```

```
    </label>
```

```
    <label class="checkbox-label">
```

```
      <input type="checkbox" id="ignoreCase" checked> Case Insensitive
```

```
    </label>
```



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</div>

<button type="submit" class="btn-primary">Evaluate Palindrome</button>

</form>

</section>

<section class="card">

<h2>Evaluation Result</h2>

<div id="outputDisplay">

<p class="placeholder-text">Enter text and click evaluate to view execution results.</p>

</div>

</section>

<section class="card full-width">

<h2>Encapsulated Session Analytics (Closure State)</h2>

<div class="analytics-grid">

<div class="stat-box">

0

Total Checks

</div>

<div class="stat-box">

0

Palindromes Found

</div>

<div class="stat-box">



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0

Errors Caught

</div>

</div>

<div class="history-container">

<h3>Recent Inspection Log</h3>

<ul id="historyList" class="history-list">

</div>

</section>

</div>

<script src="script.js"></script>

</body>

</html>

Styles.css

:root {

--primary: #be123c;

--primary-hover: #9f1239;

--secondary: #fb7185;

--bg: #fff1f2;

--card-bg: #ffffff;

--text: #4c0519;

--text-muted: #9f1239;

--border: #fecdd3;

--success: #15803d;



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--error: #b91c1c;

}

* { box-sizing: border-box; margin: 0; padding: 0; font-family: 'Segoe UI', sans-serif; }

body { background-color: var(--bg); color: var(--text); padding: 20px; }

.container { max-width: 1000px; margin: 0 auto; display: grid; grid-template-columns: 1fr 1fr; gap: 20px; }

.card { background: var(--card-bg); padding: 24px; border-radius: 12px; }

.full-width { grid-column: 1 / -1; }

.profile-header-card {

background: var(--primary); color: white; padding: 16px 24px; border-radius: 8px;

display: flex; justify-content: space-between; align-items: center;

}

.detail-value { font-size: 18px; font-weight: 700; background: rgba(255,255,255,0.18); padding: 4px 12px; border-radius: 4px; }

.form-group { margin-bottom: 14px; }

label { display: block; margin-bottom: 6px; font-size: 13px; font-weight: 600; }

input[type="text"] { width: 100%; padding: 10px; border: 1px solid var(--border); border-radius: 6px; }

.btn-primary { background-color: var(--primary); color: white; border: none; padding: 12px; border-radius: 6px; cursor: pointer; width: 100%; font-weight: 600; }

.result-box { padding: 16px; border-radius: 8px; border-left: 5px solid var(--border); background: #fff5f5; }

.result-box.success { border-left-color: var(--success); background: #f0fdf4; }

.result-box.error { border-left-color: var(--error); background: #fef2f2; }



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```
.analytics-grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 12px; margin-bottom: 20px; }
```

```
.stat-box { background: #fff5f5; border: 1px solid var(--border); padding: 12px; border-radius: 8px; text-align: center; }
```

```
.stat-value { display: block; font-size: 24px; font-weight: 700; color: var(--primary); }
```

```
.history-list { list-style: none; max-height: 150px; overflow-y: auto; }
```

```
.history-item { display: flex; justify-content: space-between; padding: 6px 10px; border-bottom: 1px solid var(--border); font-size: 12px; }
```

script.js

```
const STUDENT_NAME = "Parth Jani";
```

```
const STUDENT_PRN = "24070521241";
```

```
const GLOBAL_MAX_LENGTH = 100;
```

```
const createAnalyticsTracker = () => {
```

```
  let totalChecks = 0;
```

```
  let palindromeCount = 0;
```

```
  let errorCount = 0;
```

```
  const historyLog = [];
```

```
  return {
```

```
    recordSuccess: (originalText, isPalindrome, cleanedText) => {
```

```
      totalChecks++;
```

```
      if (isPalindrome) palindromeCount++;
```

```
      historyLog.unshift({
```

```
        text: originalText,
```

```
        status: isPalindrome ? 'Palindrome' : 'Not Palindrome',
```

```
        cleaned: cleanedText,
```




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```
time: new Date().toLocaleTimeString()

});

},

recordError: () => { errorCount++; },

getStats: () => ({ totalChecks, palindromeCount, errorCount, history: [...historyLog] })

};

};

const sessionTracker = createAnalyticsTracker();

(() => {

const DOM = {

form: document.getElementById('palindromeForm'),

textInput: document.getElementById('textInput'),

ignoreSpaceCb: document.getElementById('ignoreSpace'),

ignoreCaseCb: document.getElementById('ignoreCase'),

outputDisplay: document.getElementById('outputDisplay'),

statTotal: document.getElementById('statTotal'),

statPalindromes: document.getElementById('statPalindromes'),

statErrors: document.getElementById('statErrors'),

historyList: document.getElementById('historyList')

};

function sanitizeText(inputStr, stripPunctuation, forceLowerCase) {

let processed = inputStr;

if (forceLowerCase) processed = processed.toLowerCase();
```



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```
if (stripPunctuation) processed = processed.replace(/[^a-zA-Z0-9]/g, "");

return processed;

}

const reverseString = (str) => str.split("").reverse().join("");

const checkPalindromeLogic = function (str) {

    if (str.length === 0) return false;

    return str === reverseString(str);

};

const evaluateInput = (rawText) => {

    try {

        if (!rawText || rawText.trim().length === 0) throw new Error("Input string cannot be empty.");

        if (rawText.length > GLOBAL_MAX_LENGTH) throw new Error(`Input exceeds maximum allowed limit of ${GLOBAL_MAX_LENGTH} characters.`);

        if (/^\d+$/ .test(rawText.trim())) throw new Error("Numeric-only inputs are invalid.");

        const cleaned = sanitizeText(rawText, DOM.ignoreSpaceCb.checked, DOM.ignoreCaseCb.checked);

        if (cleaned.length === 0) throw new Error("Sanitization removed all valid characters.");

        const isPal = checkPalindromeLogic(cleaned);

        sessionTracker.recordSuccess(rawText, isPal, cleaned);

        renderSuccessResult(rawText, cleaned, isPal);

    } catch (error) {
        // Error handling logic
    }
}
```



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```
} catch (err) {

    sessionTracker.recordError();

    renderErrorResult(err.message);

} finally {

    updateAnalyticsUI();

}

};

const renderSuccessResult = (original, cleaned, isPalindrome) => {

    DOM.outputDisplay.innerHTML = `

        <div class="result-box ${isPalindrome ? 'success' : 'error'}">

            <div class="result-title" style="color: ${isPalindrome ? 'var(--success)' : 'var(--error)}">

                ${isPalindrome ? '✓ Valid Palindrome' : '✗ Not a Palindrome'}

            </div>

            <div class="result-detail"><strong>Student Evaluated:</strong>
            ${STUDENT_NAME} ${STUDENT_PRN}</div>

            <div class="result-detail"><strong>Original Input:</strong> "${original}"</div>

            <div class="result-detail"><strong>Processed Text:</strong> "${cleaned}"</div>

            <div class="result-detail"><strong>Reversed Text:</strong>
            "${reverseString(cleaned)}"</div>

        </div>

    `;

};

const renderErrorResult = (errorMessage) => {
```



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```
DOM.outputDisplay.innerHTML = `
```

```
<div class="result-box error">
```

```
<div class="result-title" style="color: var(--error)">Validation Exception  
Caught</div>
```

```
<div class="result-detail"><strong>Evaluated By:</strong> ${STUDENT_NAME}  
(${STUDENT_PRN})</div>
```

```
<div class="result-detail"><strong>Error Message:</strong>  
${errorMessage}</div>
```

```
</div>
```

```
`;
```

```
};
```

```
const updateAnalyticsUI = () => {
```

```
  const { totalChecks, palindromeCount, errorCount, history } =  
  sessionTracker.getStats();
```

```
  DOM.statTotal.textContent = totalChecks;
```

```
  DOM.statPalindromes.textContent = palindromeCount;
```

```
  DOM.statErrors.textContent = errorCount;
```

```
  DOM.historyList.innerHTML = history.map(item => `
```

```
    <li class="history-item">
```

```
      <span><strong>"${item.text}"</strong> (${item.status})</span>
```

```
      <span style="color: var(--text-muted);">${item.time}</span>
```

```
    </li>
```

```
  `).join("");
```

```
};
```

```
DOM.form.addEventListener('submit', (e) => {
```



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```
e.preventDefault();
```

```
evaluateInput(DOM.textInput.value);
```

```
});
```

```
})();
```

4. Output:

Student PRN: 24070521241

Palindrome Evaluator

Enter Phrase or Word

☒ Ignore Spaces & Punctuation
☒ Case Insensitive

Evaluate Palindrome

Evaluation Result

✗ Not a Palindrome

Original Input: "Sing"
Processed Text: "sing"
Reversed Text: "gnis"

Encapsulated Session Analytics (Closure State)

1
Total Checks

0
Palindromes Found

0
Errors Caught

Recent Inspection Log

"Sing" (Not Palindrome) 11:41:37 PM

- **File Path:**
/Users/pj/Downloads/JavaScript/Experiments/Experiment-4(24070521241)
- **Student Name:** Parth Jani
- **Student PRN:** 24070521241

5. Result/Conclusion

Diverse function implementations (IIFE, Arrow functions, declarations), scope boundaries, closures for private session states, and try...catch error handling were executed for **Parth Jani (PRN: 24070521241)**.



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Case Study-4

1. Title: Modern Multi-Theme Task Manager & Interactive To-Do Application

2. Software/Tools Required

- **Operating System:** Windows 10/11 / macOS / Linux
- **IDE/Editor:** Visual Studio Code / IntelliJ IDEA
- **Web Browser:** Google Chrome / Mozilla Firefox / Microsoft Edge
- **Languages/Tech:** HTML5, CSS3, JavaScript (ES6+)

3. Program Code:

Index.html

```
<!DOCTYPE html>

<html lang="en" data-theme="light">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Multi-Theme Task Manager</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">

  <header class="badge-header full-width">
```



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<div>STUDENT NAME: Parth Jani</div>

<div>STUDENT PRN: 24070521241</div>

</header>

<section class="card full-width flex-between">

<h2>Multi-Theme Task Manager</h2>

<div class="theme-selector">

<label for="themeSelect">Theme:</label>

<select id="themeSelect">

<option value="light">Light Mode</option>

<option value="slate">Dark Slate</option>

<option value="oceanic">Oceanic Blue</option>

</select>

</div>

</section>

<section class="card">

<h2>New Task Input</h2>

<form id="taskForm">

<div class="form-group">

<label for="taskDesc">Task Description</label>

<input type="text" id="taskDesc" placeholder="e.g. Complete Lab Practical 4 Report" required>

</div>

<div class="form-group">

<label for="priority">Priority</label>



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```
<select id="priority">

  <option value="Low">Low</option>

  <option value="Medium">Medium</option>

  <option value="High">High</option>

</select>

</div>

<button type="submit" class="btn-primary">Add Task</button>

</form>

</section>

<section class="card">

  <h2>Active Task List</h2>

  <div class="filter-bar">

    <button class="filter-btn active" data-filter="all">All</button>

    <button class="filter-btn" data-filter="active">Active</button>

    <button class="filter-btn" data-filter="completed">Completed</button>

    <button id="purgeBtn" class="purge-btn">Purge Completed</button>

  </div>

  <ul id="taskList" class="task-list"></ul>

</section>

</div>

<script src="script.js"></script>

</body>

</html>
```

Style.css



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```
:root[data-theme="light"] { --bg: #f8fafc; --card: #ffffff; --text: #0f172a; --primary: #2563eb; }
```

```
:root[data-theme="slate"] { --bg: #0f172a; --card: #1e293b; --text: #f8fafc; --primary: #38bdf8; }
```

```
:root[data-theme="oceanic"] { --bg: #042f2e; --card: #115e59; --text: #f0fdf4; --primary: #2dd4bf; }
```

```
* { box-sizing: border-box; margin: 0; padding: 0; font-family: 'Segoe UI', sans-serif; }
```

```
body { background: var(--bg); color: var(--text); padding: 20px; transition: all 0.3s; }
```

```
.container { max-width: 900px; margin: 0 auto; display: grid; grid-template-columns: 1fr 1fr; gap: 20px; }
```

```
.full-width { grid-column: 1 / -1; }
```

```
.badge-header { background: var(--primary); color: white; padding: 14px 20px; border-radius: 8px; display: flex; justify-content: space-between; }
```

```
.card { background: var(--card); border: 1px solid rgba(0,0,0,0.1); padding: 20px; border-radius: 10px; }
```

```
.flex-between { display: flex; justify-content: space-between; align-items: center; }
```

```
.form-group { margin-bottom: 12px; }
```

```
label { display: block; font-size: 12px; font-weight: 600; margin-bottom: 4px; }
```

```
input, select { width: 100%; padding: 8px; border: 1px solid #cbd5e1; border-radius: 6px; }
```

```
.btn-primary { background: var(--primary); color: white; border: none; padding: 10px; width: 100%; border-radius: 6px; cursor: pointer; }
```

```
.filter-bar { display: flex; gap: 6px; margin-bottom: 12px; }
```

```
.filter-btn { padding: 4px 8px; border: 1px solid var(--primary); background: transparent; color: var(--text); cursor: pointer; border-radius: 4px; }
```

```
.filter-btn.active { background: var(--primary); color: white; }
```

```
.purge-btn { margin-left: auto; background: #ef4444; color: white; border: none; padding: 4px 8px; border-radius: 4px; cursor: pointer; }
```

```
.task-list { list-style: none; }
```

```
.task-item { display: flex; justify-content: space-between; padding: 8px 0; border-bottom: 1px solid #e2e8f0; }
```



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1px solid rgba(0,0,0,0.1); }

script.js

```
const TaskManager = (() => {
```

```
  let tasks = [];
```

```
  let currentFilter = 'all';
```

```
  const DOM = {
```

```
    themeSelect: document.getElementById('themeSelect'),
```

```
    form: document.getElementById('taskForm'),
```

```
    desc: document.getElementById('taskDesc'),
```

```
    priority: document.getElementById('priority'),
```

```
    list: document.getElementById('taskList'),
```

```
    filterBtns: document.querySelectorAll('.filter-btn'),
```

```
    purgeBtn: document.getElementById('purgeBtn')
```

```
  };
```

```
  const render = () => {
```

```
    DOM.list.innerHTML = '';
```

```
    let filtered = tasks.filter(t => {
```

```
      if (currentFilter === 'active') return !t.completed;
```

```
      if (currentFilter === 'completed') return t.completed;
```

```
    return true;
```



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```
});
```

```
filtered.forEach((t, i) => {  
  DOM.list.innerHTML += `  
    <li class="task-item">  
      <span>  
        <input type="checkbox" ${t.completed ? 'checked' : ''}  
onchange="TaskManager.toggle(${i})">  
        <span style="${t.completed ? 'text-decoration:  
line-through;' : ''}">${t.desc} (${t.priority})</span>  
      </span>  
      <button onclick="TaskManager.remove(${i})">X</button>  
    </li>  
  `;  
});  
};
```

```
return {  
  init: () => {  
    DOM.themeSelect.addEventListener('change', (e) => {  
      document.documentElement.setAttribute('data-theme',  
e.target.value);  
    });  
    DOM.form.addEventListener('submit', (e) => {
```



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```
e.preventDefault();

tasks.push({ desc: DOM.desc.value.trim(), priority:
DOM.priority.value, completed: false });

DOM.desc.value = "";

render();

});

DOM.filterBtns.forEach(btn => {

  btn.addEventListener('click', (e) => {

    DOM.filterBtns.forEach(b => b.classList.remove('active'));

    e.target.classList.add('active');

    currentFilter = e.target.dataset.filter;

    render();

  });

});

DOM.purgeBtn.addEventListener('click', () => {

  tasks = tasks.filter(t => !t.completed);

  render();

});

},

toggle: (i) => { tasks[i].completed = !tasks[i].completed; render(); },

remove: (i) => { tasks.splice(i, 1); render(); }

};

})();
```



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```
document.addEventListener('DOMContentLoaded', TaskManager.init);
```

4. Output:

Student PRN: 24070521241

Task Lifecycle Manager

Active UI Theme: Dark Slate

Create New Task

Task Description

e.g. Complete Lab Practical 7

Priority Level

High Priority

Add Task

Task List

All Active Completed

Complete Lab practical 4

Delete

1 task remaining

Purge Completed

- **File Path:**
/Users/pj/Downloads/JavaScript/Case-Study/CaseStudy-4
- **Student Name:** Parth Jani
- **Student PRN:** 24070521241



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